



ESPRIT · 2026

FINAL YEAR PROJECT



AI-Powered Recruitment Platform

From job discovery to evidence-based hiring decisions.

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01 / 23 · INTRODUCTION

The screenshot displays the PEAXIS recruitment platform interface. At the top, a browser tab shows 'northwind.peaxis.com/pipeline'. The main view is titled 'Backend Engineer' with a subtitle 'Pipeline · 6 candidates'. It is divided into two columns: 'Applied' and 'Screening', each with a count of 2. The 'Applied' column lists two candidates: Karim Haddad (Full-stack developer, Alignment 62%) and Omar Chaabane (Backend developer, Alignment 71%). The 'Screening' column lists two candidates: Amina Benali (Backend engineer, Alignment 85%) and Sarah Mansour (DevOps engineer, Alignment 78%, marked as 'Reviewed'). A modal window is open for Amina Benali, showing her profile (Backend engineer · 4 years · Tunis) and a score of 85%. Below the profile, a section titled 'REQUIREMENTS · LINKED TO CV EVIDENCE' lists five items: NestJS (Satisfied), PostgreSQL (Satisfied), Docker (Satisfied), 3+ years backend (Satisfied), and Kubernetes (Needs verification). At the bottom of the modal are three buttons: 'Schedule interview', 'Move stage', and 'Reject'. A separate overlay shows a circular progress indicator for 'Alignment' at 85%, with the text 'traced to CV evidence'.



CONTEXT

Built inside **Prospecter**, an AI product company

THE HOST COMPANY

AI-powered B2B platform: autonomous AI SDRs for outbound prospecting.

 LLM workflows

 Multi-tenant SaaS

 Background workers

MY ROLE: THREE HATS

 **Software engineering**
Web, API, business rules

 **AI engineering**
LLM integration, evidence assessment

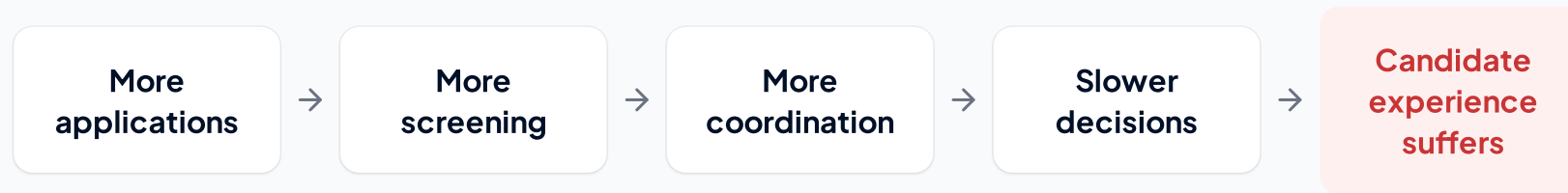
 **Platform engineering**
Deployment, security, operations

An independent product, from architecture and implementation through production deployment.



PROBLEM

Recruitment in the digital era



≈ 1.5 months median time-to-fill, for executive and non-executive roles alike (SHRM, 2025)

37% of recruiting organizations are already integrating or experimenting with generative AI (LinkedIn, 2025).

AI can remove repetitive work, but hiring also demands trust, transparency and human oversight.



PROBLEM

Three core challenges



MANUAL WORKLOAD

Screening, organizing applications, interviews and follow-up.



FRAGMENTED WORKFLOWS

Jobs, spreadsheets, email and calendars live in different tools.



DECISION CLARITY

Keyword matching and opaque AI scores hide why a candidate is recommended.

| Less effort is not enough: recruiters must also understand and trust the result.



PROBLEM

Everyone optimizes something

AI is no longer rare: major ATS vendors already ship it.



JOB PLATFORMS

Reach



ESTABLISHED ATS

Process



AI RECRUITING TOOLS

Speed



PEAXIS

Evidence

One candidate-to-recruiter journey, where every AI result is traceable and the human decides.



PROBLEM · PROJECT OBJECTIVE

How can AI make recruitment more efficient without turning hiring into a black box?



Centralize the
recruitment workflow



Understand candidate
evidence



Assist recruiters with AI



Keep humans in control



METHOD

Built like a **product**, not a prototype



Backlog · sprints · incremental delivery · continuous validation

SIX QUALITY REQUIREMENTS FROM DAY ONE





METHOD & REQUIREMENTS

What each user **needs to do**

Candidate

- ✓ Discover jobs
- ✓ Build a profile from a CV
- ✓ Apply in a few clicks
- ✓ Follow every application

Recruiter

- ✓ Create and publish jobs
- ✓ Manage candidates
- ✓ Run the hiring pipeline
- ✓ Review evidence
- ✓ Schedule interviews
- ✓ Use AI assistance

Platform

- ✓ Multi-tenant organizations
- ✓ Roles and access
- ✓ Billing and entitlements
- ✓ Protected files
- ✓ Notifications



SOLUTION

PEAXIS: two experiences, one platform



PEAXIS Jobs

For candidates

Discover jobs · profile from CV · apply · track



PEAXIS Hire

For hiring teams

Jobs · pipeline · evidence · interviews · AI assistance

SHARED PLATFORM

Identity

Organizations

Permissions

Files

Billing

AI



SOLUTION

PEAXIS Jobs: the candidate experience

1 · Discover



2 · Job details



3 · Apply with CV



4 · Track application

peaxis.com/jobs

My applications

N

Backend Engineer
Northwind · Applied today

Under review

✓

Applied

2

Screening

3

Interview

4

Offer

L

DevOps Engineer
Lumen Labs · Applied 5 days ago

Interview



SOLUTION

PEAXIS Hire: the recruiter workspace

northwind.peaxis.com/pipeline

Backend Engineer

Pipeline · 6 candidates

KanbanList

Applied2

KH

Karim Haddad
Full-stack developer

Alignment **62%**

OC

Omar Chaabane
Backend developer

Alignment **71%**

Screening1

SM

Sarah Mansour
DevOps engineer

Alignment **78%** ✓ Reviewed

Interview2

LG

Lina Gharbi
Backend engineer

Alignment **90%** ✓ Reviewed

AB

Amina Benali
Backend engineer · 4 yrs

Alignment **85%**

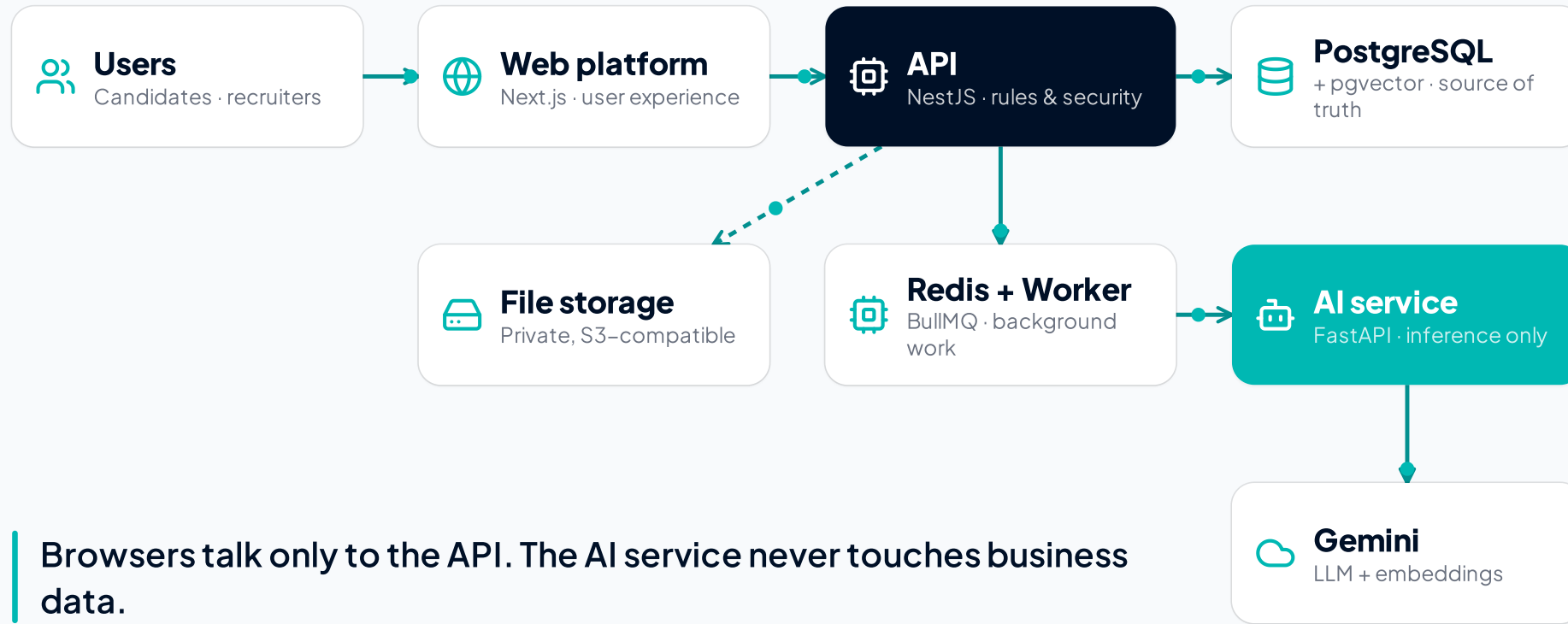
Interview · Thu 10:00

Offer0



ENGINEERING

How the pieces fit together



Browsers talk only to the API. The AI service never touches business data.



ENGINEERING & AI

How PEAXIS understands a CV

AMINA BENALI

Backend Engineer · Tunis

EXPERIENCE

Backend Engineer · Northwind, 2021 – 2025

- Built REST APIs with NestJS serving 40k users
- Designed PostgreSQL schemas and migrations
- Containerized services with Docker

SKILLS

TypeScript, NestJS, PostgreSQL, Docker, CI/CD

EDUCATION

Software Engineering degree, 2021



Candidate profile

✓ Checked against source

ROLE	Backend Engineer	✓
EXPERIENCE	Northwind · 2021 – 2025 (4 years)	✓
SKILLS	<u>NestJS</u> <u>PostgreSQL</u> <u>Docker</u> <u>TypeScript</u>	✓
EDUCATION	Software Engineering · 2021	✓

Extract content



AI structures information



Validate against source



Profile & evidence

PEAXIS does not blindly trust generated output. Parsed information is checked against the source before it becomes usable evidence.



ENGINEERING & AI

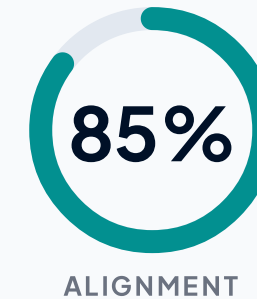
Why does this candidate match?

JOB REQUIREMENTS

CANDIDATE EVIDENCE

ASSESSMENT

NestJS Required	→	<i>"Built REST APIs with NestJS..."</i>	→	✓ Satisfied
PostgreSQL Required	→	<i>"Designed PostgreSQL schemas..."</i>	→	✓ Satisfied
Docker Required	→	<i>"Containerized services with Docker..."</i>	→	✓ Satisfied
3+ years backend Required	→	<i>"Backend engineer, 2021 - 2025"</i>	→	✓ Satisfied
Kubernetes Nice to have	→	<i>No evidence found in the CV</i>	→	⚠ Needs verification



Every result can be traced back to evidence.

AI helps retrieve and classify relevant evidence. The platform applies the final assessment rules.



ENGINEERING & AI

Explainable, **human-controlled** AI

AI ASSISTS

Parse	Retrieve
Summarize	Recommend
Explain	Draft

HUMANS CONTROL

Requirements	Stage changes
Interviews	Rejection
Offers	Hiring decision

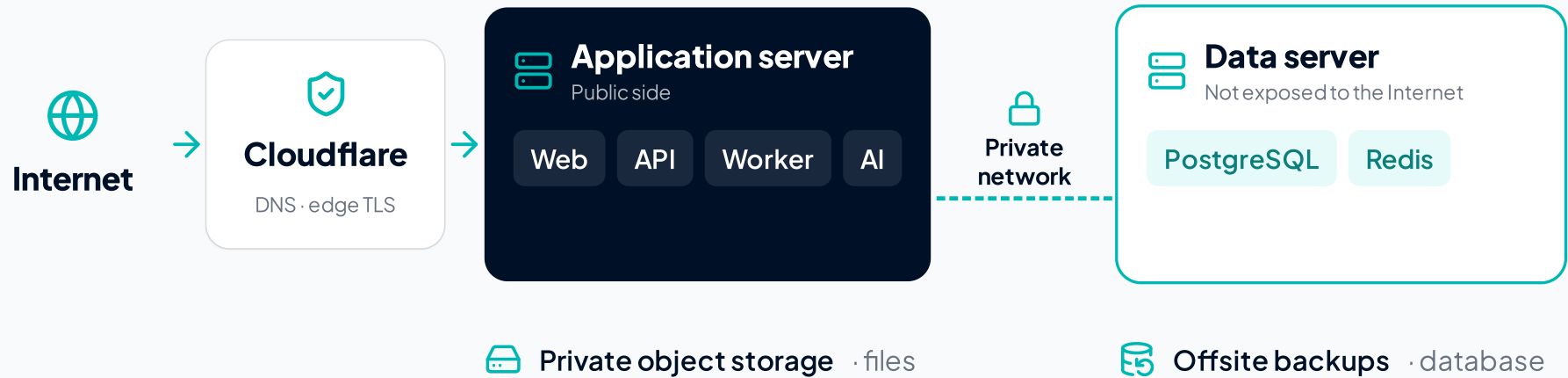
AI supports recruiter judgment. It does not replace it.

Evidence-linked · reviewable · versioned · advisory only: no AI step moves a candidate or makes an offer.



PRODUCTION

A real production deployment



The public application and the data layer are separated. The database and cache are never exposed directly to the Internet.



PRODUCTION

Engineered to be **secure, recoverable, operable**

Security

- Tenant isolation: data stays within its organization
- Secure access: hashed passwords, token sessions, MFA for platform admins
- Private data: protected files, scanned uploads
- Protected infrastructure: private database network, firewall

Reliability

- Durable background work with retries
- Events saved with the business change
- An AI outage never loses an application
- Backups with a restore procedure

Scalability

- Web, API, worker and AI are separate
- Long AI work stays off the request path
- Workers can grow independently
- Files live outside the app server

Operations

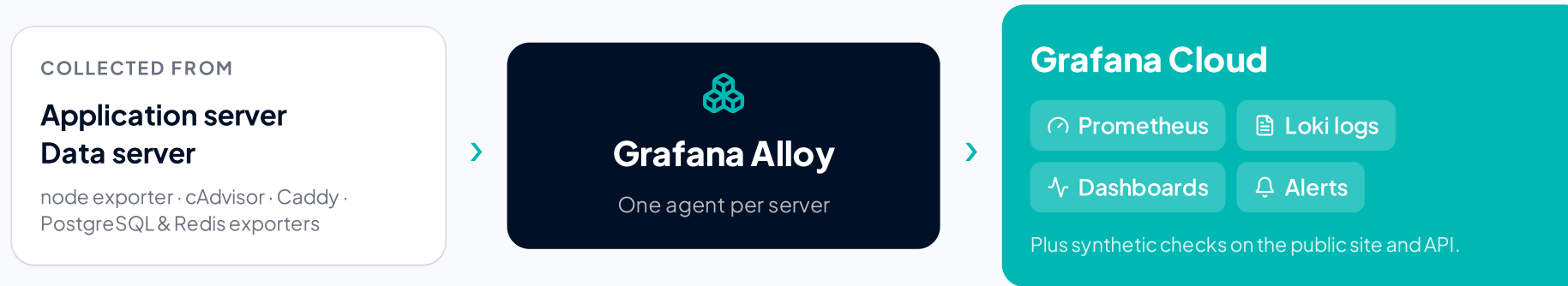
- Health checks and worker heartbeat
- Structured logs, metrics, dashboards
- Alerting through Grafana Cloud
- Next hardening: routine restore drills

We did not only build features: we designed how PEAXIS stays secure, recoverable and operable. No Kubernetes needed at today's scale; clear boundaries for tomorrow's.



PRODUCTION

We can see what the system is doing



Servers & containers

CPU, memory, disk, network



API & AI service

Errors, latency, provider calls



Queue & worker

Backlog, failures, heartbeat



Data & backups

PostgreSQL, Redis, backup status

Monitoring is configured as code and bounded, so it can never block the app. Next hardening: verify alert delivery at every release.



PRODUCTION

From Final Year Project to **real product**



Real product

Candidate and recruiter workflows



Controlled AI

Evidence-based recruiter assistance



Production architecture

Multi-tenant, deployed



Real-world validation

First client companies starting to test

PEAXIS was not built only for the defense. It is a SaaS product intended for real organizations and real users.



PERSPECTIVES

Where PEAXIS goes next

DIRECTION 1

Deepen PEAXIS Hire

Collaboration

Scorecards, feedback, shared ownership

Communication hub

Candidate email and invitations in one place

Source-linked copilot

Comparisons, interview kits, evidence gaps

Enterprise readiness

Fine-grained permissions, integrations, analytics



DIRECTION 2 **FUTURE**

PEAXIS HR

Beyond hiring: the employee lifecycle. Possible areas, not commitments:

Employee records

Leave & attendance

Documents

Performance

Employee lifecycle

Today PEAXIS focuses on recruitment. First we deepen Hire into a complete ATS; then the platform can grow from candidate management to employee management.



LIVE DEMO · ABOUT 5 MINUTES

PEAXIS in action

CANDIDATE

Discover a job



Apply with CV

RECRUITER

Application received



Review candidate



Understand evidence



Continue hiring workflow

SWITCH TO THE APP · PRESS → TO RETURN



CONCLUSION

Three ideas to remember



One platform

for the recruitment journey



Explainable AI

that assists rather than replaces
recruiters



Production engineering

built for real organizations and real
users

What started as a Final Year Project has evolved into a production SaaS platform designed to solve a real recruitment problem.



Questions?

Thank you for your attention

PEAXIS · Final Year Project · ESPRIT 2026



Production architecture in detail

EDGE

- Cloudflare is the only Internet path to the origin; firewall admits **ports 80/443 from Cloudflare ranges only**.
- Caddy terminates origin TLS and routes by host: apex and tenant subdomains → Web, admin → Admin, API → NestJS.
- Web, Admin and API bind to loopback; the worker has no port; the AI service is reachable only on the Docker network.

APPLICATION SERVER

- Containers: **web, admin, api, api-worker, ai-api, ClamAV** (malware scan).
- API and worker share one image with different commands.
- Database migration is an explicit deployment stage, never API start-up.
- Releases use immutable image digests; a manual rollback path exists.

DATA SERVER AND STORAGE

- PostgreSQL (+ pgvector) and Redis are native services bound to loopback and a **WireGuard** address.
- The firewall admits them only from the application server over WireGuard.
- Files: private S3-compatible object storage, never public.
- Backups: continuous WAL archiving plus nightly encrypted dumps to offsite object storage.



Monitoring stack in detail

COLLECTION

- **Grafana Alloy** agent on each server, configured as code in the repository.
- Host metrics (node exporter), containers (cAdvisor), Caddy metrics, PostgreSQL and Redis exporters.
- API and AI Prometheus metrics; BullMQ queue gauges and worker heartbeat; backup status file.
- Selected application, Caddy, database and Redis logs.

GRAFANA CLOUD

- Prometheus metrics and **Loki** logs, four dashboards.
- Alert rules: host and disk, worker, database and Redis, backups, API errors, ClamAV, email delivery, privacy deadlines.
- Synthetic checks on the public site and API.

SAFETY AND HONESTY

- Monitoring listeners are never public; the remote-write queue is bounded so an outage cannot block the app.
- No business identifiers are used as metric labels.
- Alert contact points and synthetic probes are re-verified at each release: the repository alone cannot prove their live state.



BACKUP · SECURITY

Security controls (verified in code and docs)

IDENTITY AND ACCESS

- Argon2 password hashes; JWT access tokens, persisted hashed refresh tokens with rotation.
- Google and LinkedIn sign-in; email verification.
- Roles per organization (owner, admin, member) mapped centrally to permissions.
- Platform admins: separate identity boundary, **TOTP MFA**, audited actions.
- Global rate limiting, tighter on sensitive public routes.

TENANT ISOLATION

- Knowing an ID never grants access: the API resolves the owning organization and checks membership.
- Employer records are read and written inside their organization context.
- Entitlements (not plan names) gate paid features.
- AI artifacts take their tenant from the application record, not from a client header.

FILES AND SERVICES

- Uploads: allowed types only, size limit, file-signature check.
- CVs stay quarantined until malware scan passes.
- Candidates see their own CV; recruiters only for applications to their organization.
- The AI service authenticates with a service secret and has no database credentials.



Evidence assessment: how the score is computed

EVIDENCE

- Recruiter-confirmed requirements per job; candidate evidence claims derived from the parsed CV.
- Direct token match first. If none, vector search over the candidate's own evidence chunks (distance ≤ 0.55 , up to 8).
- A narrow classifier only labels a **supplied** citation; it cannot invent one.

RULES (DETERMINISTIC)

- Per requirement: satisfied, partially satisfied, needs verification, unknown.
- Weights: required 0.55 · experience 0.25 · nice-to-have 0.10, normalized over the present groups.
- Transferable evidence earns 60% credit.
- A missing mandatory requirement caps the score at 59.

TRACEABILITY

- Assessments are snapshots with input fingerprint and scorer version.
- Every evaluation stores its citation, constraint check and confidence.
- Recruiters can review and waive requirements; the assessment is advisory and never changes a stage.
- Embeddings: gemini-embedding-001, 768 dimensions, pgvector.



Asynchronous processing and failure behavior

WRITE PATH

- The business transaction saves the change, its history, a versioned **Platform Event** and the AI work intent together.
- Applying never fails because Redis or the AI provider is down.

WORKER

- Durable work rows in PostgreSQL; BullMQ **ai-tasks** queue on Redis delivers them.
- A dedicated worker runs CV parsing, assessment, evidence indexing and generated content.
- Scheduled reconciliation redelivers pending or stale work; retries are bounded.

RECOVERY

- Redis loss: workers reconnect and work is rebuilt from the durable rows.
- Provider outage: applications and jobs keep working; AI artifacts show FAILED rather than silently missing.
- Notifications are idempotent: replayed events never duplicate.
- Restore runbook replays privacy tombstones before traffic resumes.



Core data model

OPERATIONAL CORE

- User · Business (tenant) · membership with role
- Job: draft, published, paused, closed, archived
- Candidate · Application with history and events
- Interview linked to an application

APPLICATION STAGES

- Applied → Screening → Interview → Offer → Hired, or Rejected
- Stage changes are made by recruiters and recorded
- Plans: Hire Free, Starter, Professional, Enterprise (entitlement-based)

DERIVED AI DATA

- Requirements, evidence claims and chunks, assessments and evaluations, CV parses, generated content, work items
- All are rebuildable and tied to their organization; stale or failed AI data never invalidates an application



Testing and delivery validation

AUTOMATED CHECKS (CI)

- Lint and typecheck; Node unit tests; Python tests for the AI service.
- Prisma validation and clean migration run against pgvector.
- API end-to-end tests with Redis; browser end-to-end tests.
- Dependency audit, secret scan (Gitleaks), image scan (Trivy).

RELEASE

- Deploy is a manual, gated workflow on a commit proven to be on main with green CI.
- Four images tagged by commit, scanned, deployed by digest.
- Backup and migration-history checks run before the schema changes.
- Post-deploy checks of public routes, headers and CORS.

HONEST LIMITS

- Hosted CI credits were unavailable for a period; releases then used local checks on the pushed commit.
- Early client validation has just begun: no usage metrics are claimed.